

TenderPulse 4IR: Commercial Executive Dossier

Next-Generation Autonomous Public Procurement Intelligence, Anti-Collusion Radar & Copernicus Sentinel-1 InSAR Spaceborne Physical Ground-Truth Platform.

SYSTEM VERSION

v4.0.0 Enterprise Production

TARGET PORTAL

eprocure.gov.bd (Bangladesh CPTU)

LEGAL FORMALISM

PPR-2008 / PPA-2006 (Rules 39, 40, 98)

RADAR PAYLOAD

Copernicus Sentinel-1 C-Band (5.405 GHz)

AUDIT SCORE

152 / 152 Pass (100%)

01

Executive Overview & Market Context

Public infrastructure procurement in emerging and rapidly industrializing economies represents over 20% to 25% of national GDP, exceeding \$15+ Billion USD annually across Bangladesh's e-GP portal alone. However, procuring authorities, international development financiers (World Bank, ADB, JICA), and honest contractors confront four systemic bottlenecks:

VECTOR 01

Cartel Ring Collusion & Cover Bidding

Syndicates coordinate artificial bid quotes and satisfy statutory 3-bidder quorum rules using shell entities, shared corporate addresses, and consecutive bank guarantees issued at identical bank branches.

VECTOR 02

Measurement Book (MB) Ghost Billing

Billions are disbursed annually for reported road carpeting or embankment reinforcement that either never occurred or washed away, without independent, verifiable physical ground-truth prior to check clearance.

VECTOR 03

Manual Prequalification Inconsistencies

High-value infrastructure bids are frequently disqualified due to minute human calculation inconsistencies in complex tender capacity equations: $A = (N \times B \times 5) - C$ or joint-venture liquidity certificates.

VECTOR 04

Commercial Balance Sheet Exposure

Contractors are forced to reveal unencumbered credit lines, tax balances, and banking agreements to review committees, exposing their commercial secrets to competitors and corrupt intermediaries.

02 Unified System Architecture

TenderPulse 4IR unites orbital Synthetic Aperture Radar (SAR), neuro-symbolic automated reasoning, graph topology analytics, and zero-knowledge cryptography into a high-density operating picture engineered to Palantir Foundry and Bloomberg Terminal standards.

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+-----+ |
TENDERPULSE 4IR ARCHITECTURE | +-----+
+-----+ ^ ^ ^ | | | +-----+ +-----+ +-----+ |
SPACE OBSERVATION| | NEURO-SYMBOLIC AI| | FORENSIC GRAPH | | Copernicus SAR | | Microsoft Z3 SMT
| | NetworkX Radar | | InSAR Coherence | | PPR-2008 Rules | | 4-Vector Rings | +-----+
+ +-----+ +-----+ ^ ^ ^ +-----+
```

```
-----+ | +-----+ | FASTAPI ENTERPRISE CORE ENGINE | |
SQLAlchemy 2.0 / PostgreSQL | | Cryptographic JWT & 3-Tier RBAC | +-----+
-----+ | +-----+ +-----+ v v v +-----+
-----+ +-----+ +-----+ | CRYPTOGRAPHIC | | AUTONOMOUS | | COMMERCIAL | |
Groth16 zk-SNARK | | 24/7 Harvester | | BankConnect | | Zero-Knowledge | | Multi-Agency | |
Credit Gates | +-----+ +-----+ +-----+
```

03

The 6 Core Defense & Intelligence Pillars

PILLAR 01

Sentinel-1 InSAR Radar

ESA Copernicus C-band SAR dual-pol microwaves penetrate monsoon clouds and night to audit civil works compaction, surface subsidence, and earthwork completion against claimed MB bills.

- ▶ InSAR phase coherence (γ) tracking
- ▶ SHA-256 disk LRU cache (85% PU savings)

PILLAR 02

4-Vector Cartel Radar

Ingests 5+ years of historical tender awards to expose syndicated price-fixing cartels and collusive cliques through NetworkX graph theory and Louvain community clustering.

- ▶ Consecutive bank guarantee tracking
- ▶ Corporate co-location & TIN matching
- ▶ Cover-bidding spread deflection

PILLAR 03

Microsoft Z3 SMT Solver

Converts the statutory text of PPR-2008 Rules 39, 40, and 98 into first-order formal logic propositions, generating deterministic SAT/UNSAT proof certificates in under 5ms.

- ▶ Capacity calculation: $A = (N \times B \times 5) - C$
- ▶ Joint-venture turnover verification

PILLAR 04

Groth16 zk-SNARK Vault

Contractors generate verifiable cryptographic solvency proofs over elliptic curve BN254, proving liquid asset thresholds without revealing private bank balances to competitors.

- ▶ Zero-Knowledge commercial privacy
- ▶ Constant-time $O(1)$ proof validation

PILLAR 05

24/7 Harvester & Corrigenda

Continuous multi-agency daemon crawls e-GP across RHD, LGED, PWD, BWDB, BREB, EED, and DGHS with token-bucket randomized jitter and proxy egress rotation.

- Corrigendum delta tracking
- Deadline extension push notifications

PILLAR 06

BankConnect Gateway

Commercial banking bridge integrated with Prime Bank, City Bank, BRAC Bank, and Eastern Bank to simulate working capital loans and automate pre-approval tracking letters.

- Automated LOC tokens (EGP-LOC-*)
- Real-time credit capacity simulation

04 Enterprise Security & 3-Tier RBAC

Authentication is enforced by cryptographic RSA-256 / HMAC-SHA256 JSON Web Tokens with standard expiration, refresh token rotation, and cryptographic JTI blacklisting.

ROLE	TARGET PERSONA	AUTHORIZED MODULES & CAPABILITIES
ROLE_EXECUTIVE	Managing Director / C-Suite	Full financial bids, Groth16 ZKP proof generation, Bank credit line pre-approvals, contract approvals.
ROLE_ANALYST	Senior Estimator / Engineer	BOQ schedule parsing, Z3 SMT legal rule solving, Tender capacity equations, Bid/No-Bid evaluation.
ROLE_AUDITOR	Anti-Corruption / Oversight	Forensic Cartel Radar inspection, Copernicus Sentinel SAR MB audits, CPTU compliance matrix, Cache clearing.
ROLE_ADMIN	Platform Superuser	Global governance, Harvester daemon execution, Redis cache controls, User registry administration.

05 Commercial Deployment Models & ROI

Model A: Tier-1 Contractors

Target: Grade-A Civil Construction Conglomerates (Mir Akhter, Spectra, Abdul Monem).

- ▶ **0% Prequalification Errors:** SMT solver eliminates clerical bid rejections.
- ▶ **40+ Hours Saved:** Instant PDF-to-BOQ extraction per tender.
- ▶ **Commercial Privacy:** Balance sheets masked via zk-SNARKs.

Model B: Oversight & Donors

Target: CPTU, IMED, Anti-Corruption Commission (ACC), World Bank, Asian Development Bank.

- ▶ **Collusion Ring Interception:** Exposes syndicate cover bidders across 5 years of data.
- ▶ **Ghost Billing Elimination:** Spaceborne SAR verifies ground truth before disbursements.
- ▶ **Automated Audit Trails:** Instant PDF/JSON legal compliance logs.

Model C: Infrastructure Banks

Target: Commercial Banks, Export Credit Agencies, Infrastructure Funds.

- ▶ **Milestone Verification:** Orbital radar audits ensure physical progress prior to loan tranche releases.
- ▶ **Contractor Pipeline:** Instant liquidity assessment for pre-approving bank guarantees.

06

REST API Integration Reference

The platform exposes a high-performance REST API operating on port 8080, protected by OAuth2 Bearer JWT authorization:

GET `/api/health`

Returns operational telemetry, uptime, and versions of all 13 active AI micro-engines.

POST /api/auth/login

Authenticates user with email/password and issues cryptographic JWT access and refresh token pair.

GET /api/tenders/live?limit=50

Queries live tenders from persistent relational store with spatial district coordinates.

POST /api/cartel/analyze

Guarded: AUDITOR, ADMIN

Executes 4-vector collusion analysis across bidding syndicates, returning Market Integrity Index and cliques.

POST /api/sentinel/query

Queries Copernicus Sentinel-1 C-band SAR rasters, 3D elevation matrices, and InSAR coherence decay.

POST /api/smt/verify

Guarded: ANALYST, EXECUTIVE, ADMIN

Executes Microsoft Z3 formal verification on PPR-2008 Rules 39/40/98, generating SAT/UNSAT proof certificates.

POST /api/zkp/prove

Guarded: EXECUTIVE, ADMIN

Generates Groth16 zero-knowledge proof proving statutory financial solvency over elliptic curve BN254.

GET /api/corrigenda

Returns statutory audit log of tender deadline extensions and security amendments detected by Harvester.

07 Automated Test Suite & Production Certification

TenderPulse 4IR has passed all 10 automated test suites (152 total validation checks) with an absolute 100% pass rate:

TEST SUITE	OPERATIONAL PILLAR	VERIFIED SPECIFICATIONS	STATUS
test_phase1_deployment.py	Containerization & Deployment	Dockerfile non-root security, Nginx Gzip reverse proxy, Compose schema.	6 / 6 PASS
test_phase2_database.py	Relational DB & GIS	SQLAlchemy 2.0 multi-mode, SQLite/Postgres persistence, BBOX queries.	3 / 3 PASS
test_phase3_security_rbac.py	Cryptographic Security & RBAC	RSA-256 JWT generation, refresh rotation, JTI blacklist, 3-tier RBAC guards.	4 / 4 PASS
test_phase4_cartel_graph.py	5-Year Historical Cartel Radar	4 forensic vectors (BGs, co-location, cover spreads, rotational reciprocity).	2 / 2 PASS
test_phase5_harvester.py	Autonomous 24/7 Harvester	Multi-agency scheduler (RHD, LGED, PWD, etc.), Corrigendum tracker.	3 / 3 PASS
test_phase6_sentinel_cache.py	Copernicus Sentinel-1 InSAR	Process API dual-pol radar evalscript, disk LRU cache, 7-day TTL credit savings.	2 / 2 PASS
test_production_readiness.py	Master Production Audit	All 7 operational pillars audited simultaneously in under 5 seconds.	7 / 7 PASS
test_sentinel_spatial.py	Spatial Resolution	Nationwide BBOX coordinate resolution across all 64 districts of Bangladesh.	PASS
test_overview_wiring.py	Executive Dashboard UI	63 reactive DOM bindings, modal dialogs, and real-time stat cards.	PASS
test_ui_wiring.py	AI Gateway Endpoints	48 end-to-end REST tests covering all 13 integrated AI micro-services.	PASS

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Compliant with Bangladesh Public Procurement Act (PPA-2006), Public Procurement Rules (PPR-2008), and Open Contracting Data Standard (OCDS).